SITTTR KERALA — REVISION 2021

CAD Lab

Course Code: 2078 **Semester: 2** **Chemical Engineering**

[Official Syllabus Reference]

Course Overview

Program	Diploma in Chemical Engineering	**Credits**	0
Course Category	Engineering Science	**Total Hours**	45 Periods
Teaching Scheme	L:0 T:0 P:3	**Assessment**	CIE & ESE (Internal/External Lab)

The CAD Lab course is designed to provide hands-on experience in Computer Aided Design and Drafting using industry-approved software (AutoCAD). It bridges the gap between manual engineering graphics and modern digital drafting standards.

Course Outcomes (COs)

CO No.	Description	Cognitive Level
CO1	Summarize Computer Aided Design.	Understanding
CO2	Illustrate the basic commands in AutoCAD.	Applying
CO3	Sketch block and Isometric 2D drawing in AutoCAD.	Applying
CO4	Illustrate 3D modelling in AutoCAD.	Applying

Module 1: Introduction to CAD

Foundations and Industry Context (3 Hours)

Introduction to AutoCAD & CAD Advantages

CAD (Computer-Aided Design) is the use of computer systems to assist in the creation, modification, analysis, or optimization of a design.

Advantages of CAD:

- **Accuracy:** Extremely high precision compared to manual drafting.
- **Speed:** Faster creation and editing of drawings.
- **Storage:** Drawings can be stored digitally and easily shared.
- **Standardization:** Easy to maintain uniform line weights, fonts, and styles.
- **Reusability:** Components (Blocks) can be reused across multiple projects.

> **Malayalam Support:** CAD [illegible].

Industry Software & Editor Interface

Commonly used CAD software in industry: AutoCAD, SolidWorks, CATIA, MicroStation, Revit.

AutoCAD Drawing Editor Components:

- **Ribbon:** Contains toolsets organized by tabs.
- **Command Line:** The primary interface for entering commands and viewing prompts.

- **Status Bar:** Controls for drawing aids like Ortho, Snap, and Grid.
- **Drawing Area:** The canvas where designs are created.
- **UCS Icon:** Shows the orientation of X and Y axes.

Module 2: Basic Commands & 2D Drafting

The Core of AutoCAD Operations (21 Hours)

Initiating a Drawing & Utility Commands

Before drawing, the environment must be set up:

- **UNITS:** Sets the measurement system (Architectural, Decimal, Engineering).
- **LIMITS:** Defines the boundaries of the drawing area.
- **ZOOM:** Controls the display magnification (Zoom All, Zoom Window).
- **SNAP & GRID:** Aids for precise positioning.
- **Function Keys:** F3 (OSNAP), F7 (GRID), F8 (ORTHO).

Setup Procedure:
1. Type `UNITS` → Set to Decimal, Precision 0.00.
2. Type `LIMITS` → Lower Left (0,0), Upper Right (420,297 for A3).
3. Type `Z` → `A` (Zoom All).

Drawing & Modify Commands

Drawing Commands:

- **LINE (L):** Creates straight line segments.
- **CIRCLE (C):** Creates circles using Center/Radius or Diameter.
- **POLYLINE (PL):** Creates a single object composed of line and arc segments.
- **HATCH (H):** Fills enclosed areas with patterns (useful for sectional views).

Modify Commands:

- **OFFSET (O):** Creates concentric circles or parallel lines.
- **TRIM (TR):** Cuts objects at a cutting edge.
- **MIRROR (MI):** Creates a reflected copy of objects.
- **FILLET (F):** Rounds the edges of objects.

> **Malayalam Support:** Modify commands □□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□ □□□□□□□□□ □□□□□□□□. □□□□□□□□□□□□, □□□□□□□□□□□□□□ □□□□□□□ □□□□□ 'Trim' □□□□□□□□□□□□□□.

Layers, Text & Dimensioning

Layers: Used to organize drawings by grouping objects (e.g., Hidden lines, Dimensions, Text). You can control color, linetype, and visibility (Freeze/Thaw).

Dimensioning: Essential for engineering drawings. Commands include DIMLINEAR, DIMANGULAR, and DIMRADIUS.

Module 3: Blocks & Isometric Drawing

Efficiency and 2D Projections (9 Hours)

Blocks and Title Blocks

A **Block** is a collection of objects that are combined into a single named object. Blocks are used for repetitive symbols (valves, pumps in Chemical Engineering).

- **BLOCK (B):** Creates a block definition.
- **INSERT (I):** Places a block into the drawing.
- **WBLOCK:** Saves a block as a separate file for use in other drawings.

Isometric Drawing

Isometric drawing is a 2D representation of a 3D object where the axes are at 120° to each other.

- **ISOPLANE:** Switches between Left, Top, and Right planes (F5 key).
- **ISOCIRCLE:** Used within the ELLIPSE command to draw circles in isometric view.

> **Malayalam Support:** Isometric drawing □□□□□□□□□□□ □□□ 3D □□□□□□□□□□ 2D □□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□□. □□□□□□□□ F5 □□ □□□□□□□□□□□ □□□□□□□□□□□□ □□□□□□□.

Module 4: 3D Modelling

Solid Modeling Fundamentals (12 Hours)

3D Workspace & Setup

To start 3D, switch the workspace to "3D Modeling". Use the **ViewCube** or **3DORBIT** to navigate.

Viewport: Divides the screen into multiple views (Top, Front, Side, Isometric) to work on 3D objects simultaneously.

3D Commands (Extrude, Presspull, Revolve)

- **EXTRUDE:** Creates a 3D solid from a 2D face by adding height.
- **PRESSPULL:** Dynamically adds or subtracts volume by selecting an enclosed area.
- **REVOLVE:** Creates a 3D solid by rotating a 2D profile around an axis (ideal for pipes/tanks).
- **3D FILLET:** Rounds the edges of a 3D solid.

Creating a Cylinder:
1. Draw a Circle in 2D.
2. Type `EXTRUDE`.
3. Select the circle and specify the height.

Coordinate Entry Guide

AutoCAD uses three main methods for entering coordinates:

```
Absolute: (X, Y)
Example: 10,20
```

```
Relative: @X, Y
Example: @50,0
```

```
Polar: @Distance < Angle
Example: @100<45
```

Diagram Library

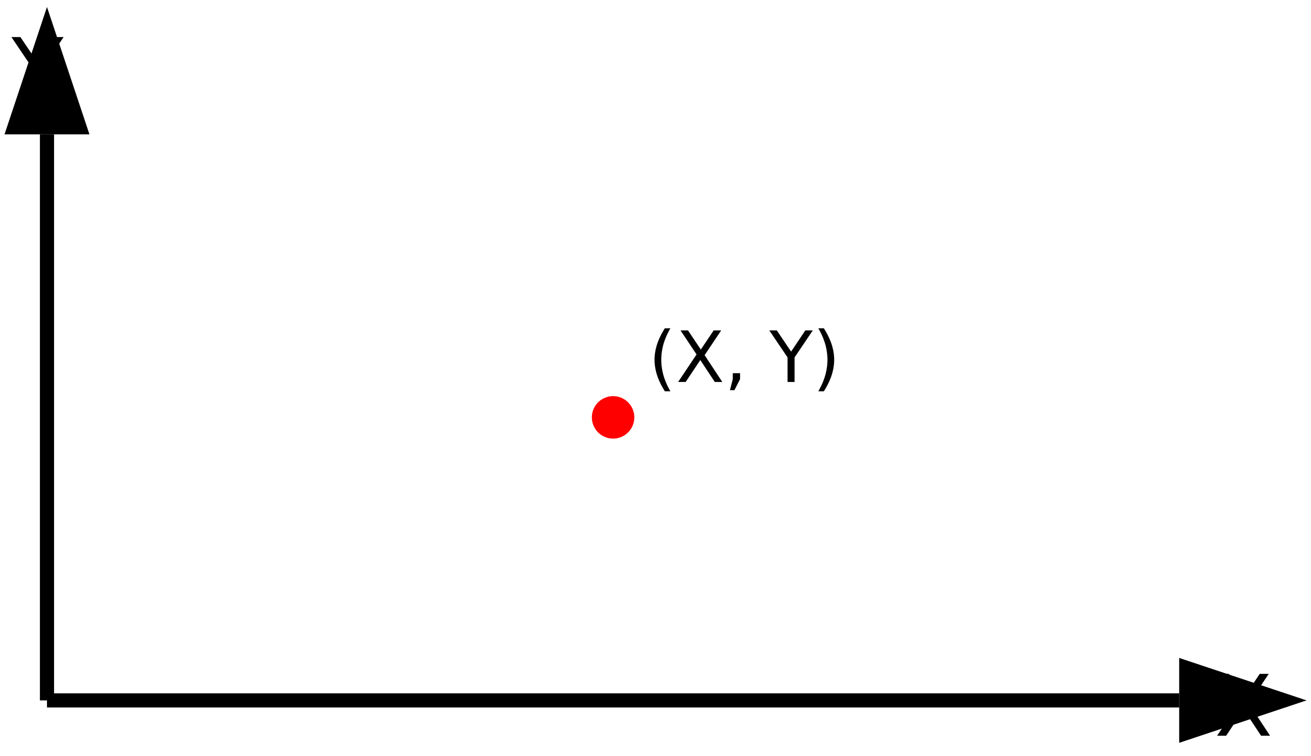

Figure 1: Cartesian Coordinate System in AutoCAD

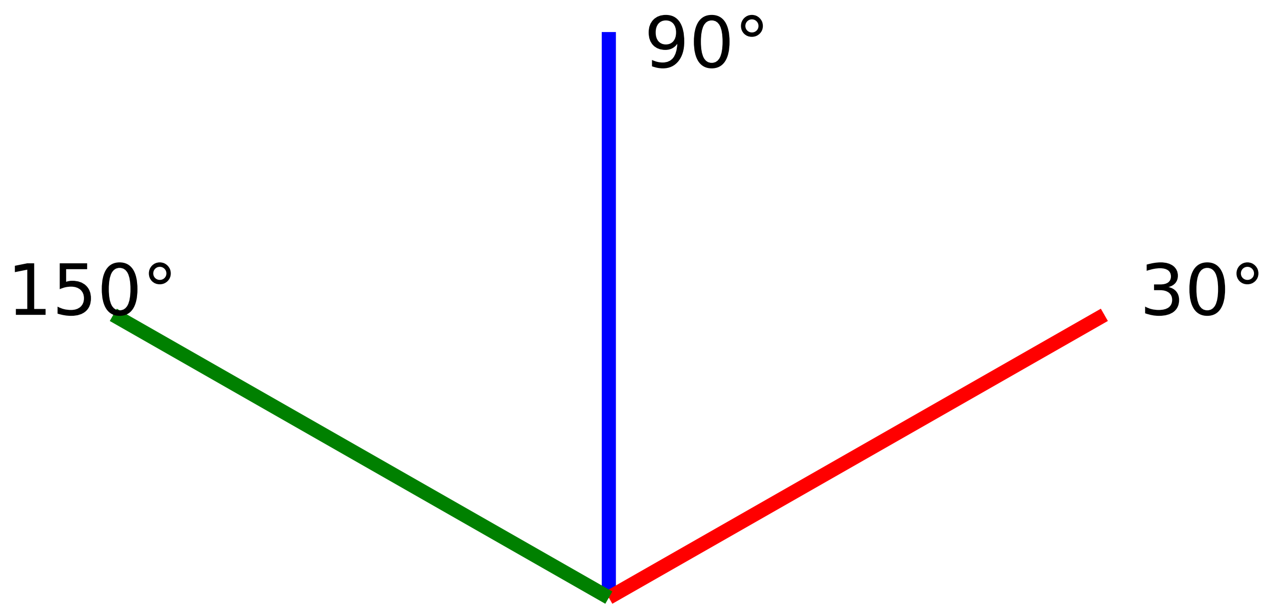

Figure 2: Isometric Axis Orientation

Practice Questions & Answers

What is the difference between 'Freeze' and 'Off' in Layers?

Off: Makes the layer invisible but AutoCAD still calculates the geometry during regenerations.
Freeze: Makes the layer invisible and AutoCAD ignores it during regenerations, improving performance in large drawings.

How do you draw a circle in an Isometric plane?

1. Ensure `ISODRAFT` is ON.
2. Type `ELLIPSE` (EL).
3. Choose the `Isocircle` option.
4. Specify center and radius.

Explain the 'Presspull' command.

Presspull is used to create or modify 3D solids. By clicking inside a bounded 2D area, you can pull it to add volume or push it to create a hole/subtract volume.

Practice Model Paper

Time: 3 Hours | Max Marks: 100

1. **Part A (Basic):** Set up an A4 sheet with decimal units and 0.0 precision. Draw a title block at the bottom right. (20 Marks)
2. **Part B (2D):** Draw the sectional front view of a Flanged Pipe Joint using appropriate layers and hatching. (40 Marks)
3. **Part C (3D):** Create a 3D model of a simple Storage Tank with a conical roof using REVOLVE or EXTRUDE commands. (40 Marks)

Quick Revision: Keyboard Shortcuts

L	Line	**CO**	Copy
C	Circle	**M**	Move
TR	Trim	**RO**	Rotate
O	Offset	**SC**	Scale
F	Fillet	**EX**	Extend
H	Hatch	**B**	Block

Glossary

- **GUI:** Graphical User Interface.
- **Ortho Mode:** Restricts cursor movement to horizontal and vertical directions.
- **OSNAP:** Object Snap (snaps to endpoints, midpoints, etc.).
- **UCS:** User Coordinate System.
- **WCS:** World Coordinate System (Fixed).

Official Source Declaration

This lesson is based on the official syllabus for **Course Code 2078 (CAD Lab)** under the **Revision 2021** curriculum provided by **SITTTR Kerala**.

Source URL: [SITTTR Syllabus Portal]

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> 2. **[illegible]:** [illegible] [illegible] 3D [illegible] 2D-[illegible] [illegible].
> 3. **3D [illegible]:** [illegible] (Extrude), [illegible] (Revolve) [illegible] [illegible] [illegible] 3D [illegible] [illegible].
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